Project proposal

MetriQ automatic inspection station

Proposal no.	P-[YYYY-NNN] rev [0]	**Date**	[DD/MM/YYYY]
Customer	[Company]	**Plant / line**	[Plant], [line]
Contact	[Name, title]	**Quotation**	Q-[YYYY-NNN]

1. Summary

Problem	[Defects escaping / manual inspection cost]
Solution	MetriQ [Pro / Lite] inspection station with automatic reject
Products	[SKUs]
Rate	[pcs/min] at [m/min]
Acceptance	Escape ≤ [x]%, false reject ≤ [y]% (§5)
Price	[VND] excl. VAT, installed (Quotation Q-[YYYY-NNN])
SAT	Week [n] after PO and deposit
Payback	[n] months: [inspectors replaced × monthly wage] or [escape cost avoided]

2. Requirements

Agreed at site visit on [date]. Source: requirements form RF-[YYYY-NNN].

#	Requirement	Target	Test method	Owner	Status
R1	Rate	[pcs/min]	Count over 30 min at FAT and SAT		
R2	Defect: [name]	Escape ≤ [x]%	[n] labelled defective samples		
R3	False reject	≤ [y]%	[n] labelled good samples		
R4	Reject output	Flagged product removed	[n] flagged products		
R5	Data	[counts, defects per shift, format]	Report generated at FAT		
R6					

Status: Agreed / Open.

3. Solution

The station mounts over the existing conveyor. The camera captures each product, MetriQ classifies it in real time, the I/O module fires the reject device, and the screen shows counts and defects per shift.

Part	Model	Supplier (Vietnam)	Qty
Camera			
Lens			
Lighting			
Edge computer	[Pro: GPU / Lite: CPU]		

Part	Model	Supplier (Vietnam)	Qty
Trigger sensor / encoder			
I/O module			
Reject device	[Air jet / pusher / existing]		
Enclosure and frame	[IP rating, material]		
Touchscreen HMI			
Power supply, UPS			

Software	MetriQ [version]. Recipe per SKU. Zero-shot setup of new defect types.
Access levels	Operator · Maintenance · Engineer
HMI language	Vietnamese and English

4. Inspection scope

Defect	Class	Smallest size (mm)	Action
	Critical		Reject
	Major		Reject
	Minor		[Reject / log]

Out of scope	[Defects, SKUs, product faces not visible to the camera]

5. Performance and acceptance

Metric	Target	Sample	Conditions
Escape rate, critical	[ ]%	[n] pcs	[speed, lighting]
Escape rate, major	[ ]%	[n] pcs	
False-reject rate	[ ]%	[n] pcs	
Rate	[pcs/min]	30 min run	
Uptime during SAT run	[ ]%	[n] h run	

FAT	At MTR workshop, [location]. Sealed validation samples on the final configuration. Customer witnesses on site or by video.
SAT	On the customer line. [n]-hour production run per SKU. Pass = all targets met. One retest included. Failures from out-of-scope conditions do not count.
Definitions	Escape rate = defective accepted ÷ defective tested. False-reject rate = good rejected ÷ good tested. Reference labels: customer QA.

6. Layout and interfaces

Layout drawing	[Drawing no., rev]. Approved by customer before parts are bought.
Remote support	[Customer-approved tool / none]
Image storage	[Station / customer server], [n] days

Signal / data	Direction	Type	Notes
Reject	MetriQ → reject device	24 V digital	
Fault / stop line	MetriQ → PLC	24 V digital	
Line running	PLC → MetriQ	24 V digital	
Shift report	MetriQ → [server / email]	CSV / PDF	

7. Scope split

Task	MTR	Customer
Design, parts, build, software	✓	
Sample collection	✓	✓
Sample labelling (reference)		✓
Layout approval		✓
Power, air, network to station point		✓
Delivery, installation, commissioning	✓	
PLC changes on existing line	[ ]	[ ]
Line stop for installation ([n] h)		✓
Training	✓	
FAT and SAT sign-off		✓

8. Plan

Week after PO	Milestone
0	PO and deposit, kick-off
1	Layout and design approved
1–3	Parts bought locally, station built, set up on samples
[4]	FAT
[5]	Delivery and installation
[6]	SAT, training, handover

Customer inputs	Layout approval within 3 working days · samples by week 1 · line access on install day. Late inputs move the plan by the same time.

9. Training and handover

Operators	[n] sessions × [n] h, each shift
Maintenance / engineer	[n] h
Handed over at SAT	User manual (Vietnamese), wiring diagram, BOM with serials, supplier documents, electrical test record, FAT/SAT records, software backup

10. Warranty and support

Warranty	12 months from SAT. Parts and labour for defects in MTR supply. Consumables excluded.
Response	Remote within [n] working hours · on site within [n] working days ([city])
After warranty	Support plan, see quotation

11. Assumptions and deviations

#	Item	Assumption or deviation
A1		
A2		
A3		

12. Approval

	Customer	MTRobotix
Name		
Title		
Signature		
Date		